Sentiment Analysis on Youtube comments

Motivation

- → Text data is a **hot topic** and **demanding** right now.
- → **Curiosity** about the **complexity** of implementing LLM architectures.
- → Interested in **challenges** like ambiguity and context in language.

Data

Source: 🤗 *Hugging Face - YouTube Comments Sentiment Analysis Dataset*

➔ **Original data set**

> 1 M of YouTube comments, each labeled with a sentiment category: Positive, Neutral, or Negative. Covering a **wide variety** of topics from programming and news to sports and politics.

➔ **Features (12)**

Such as: Video Title, Author Name, Comment Text, Likes, Replies, CountryCode, Sentiment *(Generated Manual + AI-Gemini)*.

➔ **Final Dataset**

100 K random sample, features: Comment text and Sentiment label.

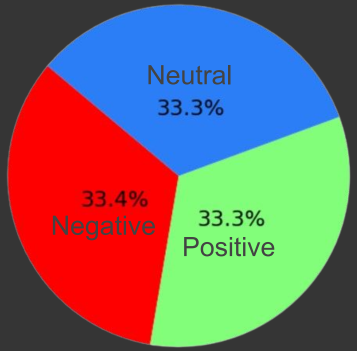
Data overview

Average length (words) per comment **19**

CommentText string	Sentiment string
Anyone know what movie this is?	Neutral
The fact they're holding each other back while equally being most aggressive 😂 😂	Positive
waiting next video will be?	Neutral

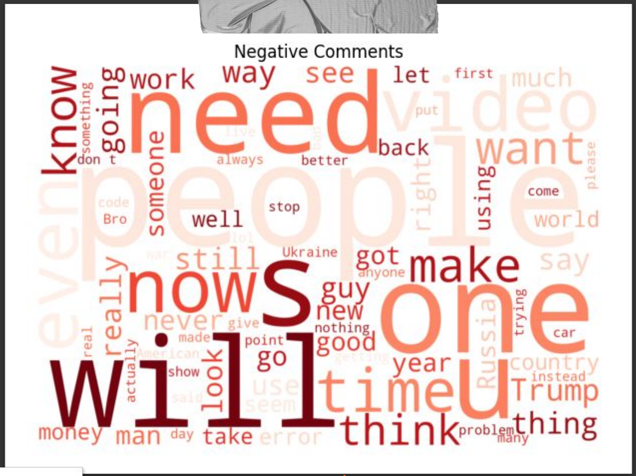
22 17 19

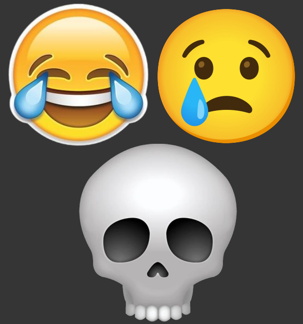
Well balance classes

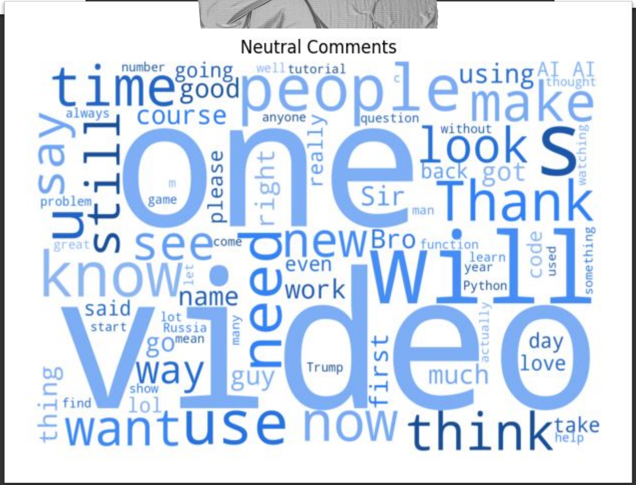


12% of the comments in languages different than English


Positive Comments
Thank
video
love
good
best
much
one
people
will
great
way
time
make
know
well
still
made
thing
Sir
want
work
back
now
man
need
awesome
see
understand
God bless
learn
tutorial
better
lot
keep
say
day
hope
really
channel
first


Negative Comments
people
need
video
will
one
know
nows
time
think
want
back
work
way
see
let
first
much
going
someone
well
stop
make
got
Ukraine
guy
new
good
year
Trump
thing
country
money
man
take
error
world
come
say




Neutral Comments
one
video
people
time
will
Thank
know
see
new
need
still
make
look
use
want
now
think
course
good
going
tutorial
using
AI
say
Sir
Bro
name
work
way
guy
first
much
day
love
take
Python
function
code

Data preprocessing

➔ **Translation**

1. Language detection with **FastText** *(returns language and confidence).*
2. Translation with **Google API**

➔ **Quick data cleanup**

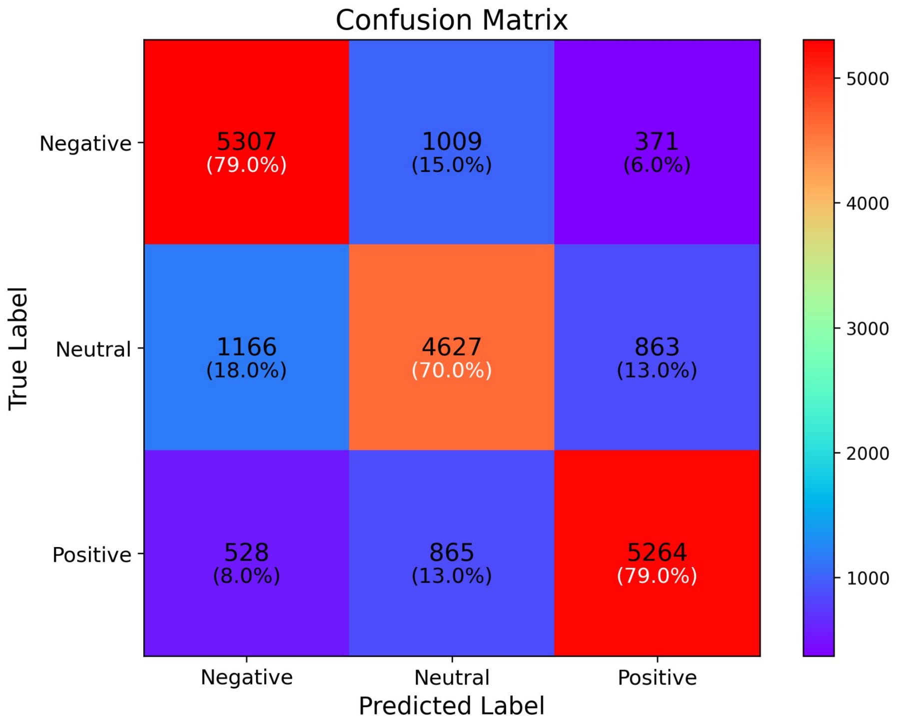
1. Remove URLs
2. Convert emojis to text (❤️ -> “heart”)
3. Remove extra spaces

Model

- We applied a **pretrained RoBERTa-base model**, originally trained on **124M English tweets** (2018–2021) and fine-tuned for sentiment analysis (Positive, Negative, Neutral).

- To adapt the model to our **YT comments dataset**, we **fine-tuned it further**, freezing the **first 4 encoder layers**. This helped reduce overfitting and **speed up training**.

- Because the model had already been trained on **social media** content, it was better able to **understand** the **variety of language** used in YouTube comments.



Test Accuracy: **76%**
F1 Score: **76%**

	Precision	Recall
Negative	0.76	0.79
Neutral	0.71	0.70
Positive	0.81	0.79

Testing

The model performs **well** in **general.**

Misclassified	True	Predicted
imma say one word: scratches	⊖	◯
People who dislike this video can't grasp the concept lol!	⊕	◯
thank you free bootcamp. please reduce your add count in this video saw more than 7 adds in just 20 minutes.	◯	⊖

😓 **Weaknesses**

- Struggles capturing complexity of **ironic or indirect expressions**.
- Difficulty in understanding **context** on longer comments
- **Recall** for Neutral (70%) is the lowest the model might be **biased** towards predicting more **extreme sentiments**

Examples

Speechless!!!!

73% 18% 9%

🏆

VS

Speechless!!!! ❤️

0% 1% 99%

Thanks for mention the worst models

72% 2% 26%

🏆

VS

Thanks for mention the worst models !

51% 2% 46%

🏆

At least investigate!

10% 65% 25%

🏆

What is your secret?? I can't beat them in 20 minutes!

46.67% 46.68% 0%

🏆

Conclusions and key insights

- **Data Preparation**: Transformers require less data prep than simpler models, but still need careful preprocessing.
- **LLMs Challenges**: Struggle with sarcasm, slang, and ambiguity in text. Specially on social media where language evolves rapidly and new terms are incorporated daily.
- **Resource Demands**: Transformers are efficient but require significant computational resources.

Thank you!